Non-Functional Requirements

Project: Real-Time IoT Data Pipeline **Team Leader:** Rana Shaaban Abdelqader Mohamed
Section: 3. Requirements Gathering

1. Purpose

This document defines how well the system must perform, not only what it must do. It covers performance, reliability, security, usability, and other quality criteria, plus the constraints and assumptions the project works under. Where possible, each requirement is measurable so it can be tested during the final demo.

Priority key: M = Must have, **S** = Should have, **C** = Could have.

2. Performance

ID	Requirement	Measure	Priority	Related FR
NFR-01	The simulator shall generate one reading every 5 seconds	Interval between readings is 5 seconds	M	FR-02
NFR-02	An alert shall be raised within 5 seconds of a threshold breach	Time from reading timestamp to alert timestamp is 5 seconds or less	M	FR-22
NFR-03	The dashboard shall show new alerts and refreshed trends without manual reload	Dashboard updates automatically within a few seconds of new data	M	FR-29, FR-31
NFR-04	The batch ETL job shall finish a batch within a time agreed by the team so the dashboard data stays reasonably current	ETL run time is recorded and reported in the final report	S	FR-16

3. Reliability and Data Quality

ID	Requirement	Measure	Priority	Related FR
NFR-05	At least 99% of generated readings shall reach Snowflake	Completeness of rows loaded vs. readings generated is 99% or more	M	FR-16, FR-18
NFR-06	No reading accepted by the API shall be lost between the API and Kafka during a normal run	Count in Kafka topic equals count accepted by the API	M	FR-09
NFR-07	A failure in one component shall not stop the other path (batch or streaming)	Stopping the batch consumer does not stop alerts, and the reverse	S	FR-10
NFR-08	Malformed or invalid readings shall not crash any component	Components log and skip bad records and keep running	S	FR-08, FR-17, FR-25
NFR-09	If Kafka setup fails, the system shall have a fallback to file-based ingestion	Documented fallback tested at least once	C	FR-09

4. Usability

ID	Requirement	Measure	Priority	Related FR
NFR-10	The dashboard shall be understandable by an operator without training	A new viewer can find current trends, anomaly count, and alert history without help	M	FR-27 to FR-30
NFR-11	Charts and tables shall have clear titles, axis labels, and units (for example degrees Celsius and percent)	All dashboard visuals labeled	M	FR-27

ID	Requirement	Measure	Priority	Related FR
NFR-12	Alerts shall be easy to notice and shall state which sensor, which metric, and what value caused them	Alert text includes sensor ID, metric, value, and threshold	M	FR-22, FR-29
NFR-13	Empty and error states shall show a clear message	Message shown when there is no data or Snowflake is unreachable	S	FR-33

5. Security

Production-grade security is out of scope. The project applies basic good practice only.

ID	Requirement	Measure	Priority
NFR-14	Snowflake credentials and other secrets shall not be stored in the code or committed to GitHub	Secrets are kept in environment variables or a git-ignored config file	M
NFR-15	The API shall accept only readings that match the schema	Invalid input is rejected	S
NFR-16	Components shall run with the minimum access they need (for example a Snowflake role limited to the project database)	Role permissions reviewed by the ETL owner	C

6. Maintainability and Portability

ID	Requirement	Measure	Priority
NFR-17	Each component shall be a separate module with a defined interface so members can build in parallel	One module per member, interfaces documented	M

ID	Requirement	Measure	Priority
NFR-18	The data schema shall be documented and frozen in Week 1; changes need team leader approval	Schema document in the repository, change log kept	M
NFR-19	Kafka shall run from a documented Docker setup so any team member can reproduce the environment	Setup notes allow a clean machine to run the pipeline	M
NFR-20	Code shall follow a consistent style and be stored in Git with meaningful commits	Repository history reviewed before final submission	S
NFR-21	Threshold values and connection settings shall be configurable without code changes	Values live in a config file or environment variables	S

7. Documentation and Cost

ID	Requirement	Measure	Priority
NFR-22	Every deliverable shall be documented as listed in the proposal timeline	Weekly documents present in GitHub	M
NFR-23	The project shall run using open-source or free tools only, with Snowflake on a trial account	Direct cost is close to zero	M

8. Project Constraints

These are fixed limits that the requirements must fit inside.

ID	Constraint	Type
C-01	The project must be completed within 4 weeks, following the weekly phases	Time

ID	Constraint	Type
C-02	Data is simulated only; no physical IoT hardware is used	Scope
C-03	Only two measurements are collected: temperature and humidity, plus sensor ID, timestamp, and location	Scope
C-04	The architecture must include both a batch path and a streaming path fed from Kafka	Course requirement
C-05	All documents and code must be uploaded to GitHub	Course requirement
C-06	The team has five members, each owning one component	Resource
C-07	Tools are limited to Python, Pandas, Flask, Apache Kafka, Snowflake (trial), Streamlit, and Git/GitHub	Technology
C-08	The budget is effectively zero; the main cost is the team's time	Cost

9. Assumptions and Dependencies

ID	Assumption or Dependency
A-01	The Snowflake trial account remains available for the whole 4 weeks
A-02	Every team member has a machine that can run Kafka (through Docker) and Python
A-03	The data schema is agreed in Week 1 and stays stable
A-04	Each component's owner delivers on time, and sample files are used to work in parallel when a dependency is late
A-05	Alert thresholds will be agreed by the team before the streaming path is built

10. How Each Requirement Will Be Verified

Area	Verification Method
Performance (NFR-01 to NFR-04)	Compare timestamps in the logs and measure alert delay during the demo
Reliability (NFR-05 to NFR-09)	Compare record counts at each stage, run the data validation report, and stop one path to test isolation
Usability (NFR-10 to NFR-13)	Dashboard walkthrough with a sample user
Security (NFR-14 to NFR-16)	Repository review for committed secrets and schema rejection tests
Maintainability (NFR-17 to NFR-21)	Fresh setup test from the documentation and repository review
Documentation and cost (NFR-22, NFR-23)	Checklist against the weekly deliverables and tool list